

Serverless computing

- is a cloud computing model where developers build and run applications without having to manage the underlying server infrastructure. Despite the name, servers are still involved—it's just that the responsibility for provisioning, scaling, and maintaining those servers is handled by the cloud provider.
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Key Concepts

1. No Server Management

- You don't need to set up, patch, or maintain servers.
- The cloud provider (like AWS, Azure, or Google Cloud) handles all the infrastructure.

2. Event-Driven Execution

- Code is typically executed in response to events, such as HTTP requests, file uploads, or database updates.
- You write small units of logic (functions) that get triggered by these events.

3. Automatic Scaling

- The platform automatically scales the application up or down based on demand.
- If there's no traffic, you aren't using resources (or paying for them).

4. Pay-as-You-Go Pricing

- You pay only for the actual compute time your code uses—measured in milliseconds—rather than for pre-allocated server capacity.
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Common Serverless Platforms

- AWS Lambda
 - Azure Functions
 - Google Cloud Functions
 - Cloudflare Workers
 - Netlify Functions
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Benefits

- **Reduced Operational Overhead:** No need to manage or configure servers.
 - **Cost-Efficiency:** Only pay for what you use.
 - **Faster Time to Market:** Focus on writing code, not infrastructure.
 - **Scalability:** Automatically scales with demand.
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Trade-Offs

- **Cold Starts:** Functions may take longer to start after being idle.
- **Limited Execution Time:** Most platforms limit how long a function can run.
- **Complex Debugging:** Harder to test and debug locally compared to traditional apps.
- **Vendor Lock-In:** You might become dependent on specific cloud provider features.

Analogy

Think of traditional computing like owning a restaurant kitchen—you buy and maintain all the equipment. Serverless is like using a food truck rental that comes fully stocked and ready to go. You just cook and serve; the vendor handles the setup, maintenance, and cleanup.
